Literature-Implied Answer: A Monotonically Sokolov Characterization

Run `fa_banach_001`

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Status. `literature_implied_answer`. This packet records a later-literature identification, not a new original proof. The source paper is Wieslaw Kubis, *Banach spaces with projectional skeletons*, arXiv:0802.1109. The supporting paper is Ondrej F. K. Kalenda, *Projectional skeletons and Markushevich bases*, arXiv:1805.11901.

Source question. In the final remarks of arXiv:0802.1109, Kubis writes that for a norming subspace $D \subset X^*$, $\mathcal{T}_D = \sigma(X, D)$, and asks:

> Find a topological property of $(X, \mathcal{T}_D)$ which says when D generates projections in X.

This appears on page 28 of the source PDF, immediately after the discussion that D generating projections implies $(X, \mathcal{T}_D)$ is Lindelof.

Supporting theorem. Kalenda's Theorem 4.1 in arXiv:1805.11901 states that, for a Banach space X and a norming subspace $D \subset X^*$, the following are equivalent, among other formulations:

$$D \text{ is induced by a projectional skeleton in } X,$$

and

$$D \text{ is weak}^*\text{-countably closed and } (X, \sigma(X, D)) \text{ is monotonically Sokolov.}$$

The theorem also gives an equivalent continuous-image formulation using monotonically Sokolov spaces.

Identification. Kubis had already proved the bridge between generating projections and induced subspaces. If a norming subspace D generates projections, then the smallest linear subspace containing D and closed under weak-star closures of countable subsets is induced by a projectional skeleton. Hence, when D itself is weak-star-countably closed, generation of projections is equivalent to D being induced by a projectional skeleton. Conversely, an induced norming subspace generates projections by Kubis's characterization of projectional skeletons via elementary submodels.

Conclusion. For the standard weak-star-countably-closed normalization of the problem, the requested topological property is monotone Sokolovness:

$$D \text{ generates projections in } X \quad \Longleftrightarrow \quad (X, \sigma(X, D)) \text{ is monotonically Sokolov,}$$

provided $D \subset X^*$ is norming and weak-star-countably closed. More precisely, the right-hand side should be read together with the weak-star-countable closedness condition on D, exactly as in Kalenda's Theorem 4.1.

Scope limitations. This does not claim a complete answer for arbitrary norming subspaces that are not weak-star-countably closed. It also does not resolve the other final questions in arXiv:0802.1109, such as the general projectional-generator problem, the closed-subspace separable-complementation problem, or the $C(\omega_2 + 1)$ embedding problem for Plichko spaces.

Search evidence. Local cheap-index searches found no previous packet for arXiv:0802.1109 or for the monotonically Sokolov characterization. Web and local source searches for exact phrases around "topological property", $\mathcal{T}_D$, $\sigma(X, D)$, "projectional generator", "generates projections", and "monotonically Sokolov" found arXiv:1805.11901 as the decisive later anchor. The relation to Kubis's exact problem is agent-identified from the two theorem statements, rather than explicitly advertised by the supporting author.

References.

- W. Kubis, *Banach spaces with projectional skeletons*, arXiv:0802.1109.
- O. F. K. Kalenda, *Projectional skeletons and Markushevich bases*, arXiv:1805.11901; Proc. London Math. Soc. (3) 120 (2020), no. 4, 514–586.